Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

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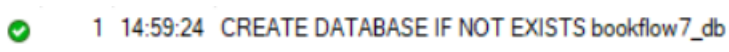
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Section – 3

Create Database:

```
CREATE DATABASE IF NOT EXISTS bookflow7_db;
```

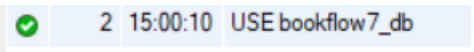
Output:



Use database:

```
USE bookflow7_db;
```

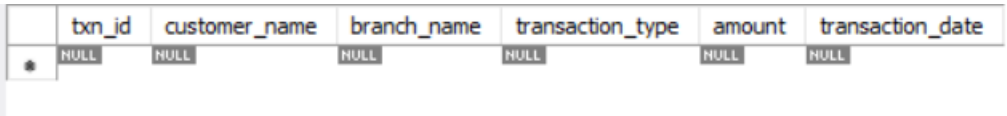
Output;



Create Table:

```
CREATE TABLE bank_transactions (
    txn_id INT PRIMARY KEY,
    customer_name VARCHAR(50),
    branch_name VARCHAR(50),
    transaction_type VARCHAR(20),
    amount DECIMAL(10,2),
    transaction_date DATE
);
```

Output;



Insert Data:

```
INSERT INTO bank_transactions VALUES
(101,'Sushma','Hyderabad','Deposit',5000,'2024-01-05'),
(102,'Sowmya','Hyderabad','Withdrawal',2000,'2024-01-06'),
(103,'Mokshagna','Vijayawada','Deposit',12000,'2024-01-08'),
(104,'Akhila','Vizag','Deposit',8000,'2024-01-10'),
(105,'Aswitha','Hyderabad','Withdrawal',3500,'2024-01-11'),
(106,'Likhitha','Vizag','Deposit',15000,'2024-01-12'),
(107,'Vaishnavi','Vijayawada','Withdrawal',1000,'2024-01-13'),
(108,'Sush','Hyderabad','Deposit',9000,'2024-01-14'),
(109,'Venugopal','Vizag','Withdrawal',4000,'2024-01-15'),
(110,'Kalavathi','Vijayawada','Deposit',11000,'2024-01-16');
SELECT * FROM bank_transactions;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Sushma	Hyderabad	Deposit	5000.00	2024-01-05
102	Sowmya	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
104	Akhila	Vizag	Deposit	8000.00	2024-01-10
105	Aswitha	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Likhitha	Vizag	Deposit	15000.00	2024-01-12
107	Vaishnavi	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Sush	Hyderabad	Deposit	9000.00	2024-01-14
109	Venugopal	Vizag	Withdrawal	4000.00	2024-01-15
110	Kalavathi	Vijayawada	Deposit	11000.00	2024-01-16
NULL	NULL	NULL	NULL	NULL	NULL

Sum():

```
SELECT SUM(amount) AS Total_Amount
FROM bank_transactions;
```

Output:

Total_Amount
70500.00

Avg():

```
SELECT AVG(amount) AS Average_Transaction
FROM bank_transactions;
```

Output:

Average_Transaction
7050.000000

Max():

```
SELECT MAX(amount) AS Highest_Transaction
FROM bank_transactions;
```

Output:

Highest_Transaction
15000.00

Min():

```
SELECT MIN(amount) AS Lowest_Transaction
FROM bank_transactions;
```

Output:

Lowest_Transaction
1000.00

Count():

```
SELECT COUNT(*) AS Total_Transactions
FROM bank_transactions;
```

Output:

Total_Transactions
10

Where + Sum():

```
SELECT SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output:

Total_Deposit
60000.00

Group By:

```
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

	branch_name	Total_Amount
▶	Hyderabad	19500.00
	Vijayawada	24000.00
	Vizag	27000.00

Group By + Order By:

```
SELECT branch_name,
       SUM(amount) AS Total_Amount
FROM bank_transactions
GROUP BY branch_name
ORDER BY Total_Amount DESC;
```

Output:

	branch_name	Total_Amount
▶	Vizag	27000.00
	Vijayawada	24000.00
	Hyderabad	19500.00

Group By + Having + Order By:

```
SELECT branch_name,
       COUNT(*) AS Total_Transactions
FROM bank_transactions
GROUP BY branch_name
HAVING COUNT(*) >= 3
ORDER BY Total_Transactions DESC;
```

Output:

	branch_name	Total_Transactions
▶	Hyderabad	4
	Vijayawada	3
	Vizag	3

Where + Count():

```
SELECT COUNT(*) AS Withdrawals
FROM bank_transactions
WHERE transaction_type='Withdrawal';
```

Output:

	Withdrawals
▶	4

Where + Avg():

```
SELECT AVG(amount) AS Avg_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit';
```

Output:

	Avg_Deposit
▶	10000.000000

Group By + Max():

```
SELECT branch_name,
       MAX(amount) AS Highest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

	branch_name	Highest_Amount
▶	Hyderabad	9000.00
	Vijayawada	12000.00
	Vizag	15000.00

Group By + Min():

```
SELECT branch_name,
       MIN(amount) AS Lowest_Amount
FROM bank_transactions
GROUP BY branch_name;
```

Output:

	branch_name	Lowest_Amount
▶	Hyderabad	2000.00
	Vijayawada	1000.00
	Vizag	4000.00

Where + Order By:

```
SELECT *
FROM bank_transactions
WHERE amount > 8000
ORDER BY amount DESC;
```

Output:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Likhitha	Likhitha	Deposit	15000.00	2024-01-12
103	Mokshagna	Vijayawada	Deposit	12000.00	2024-01-08
110	Kalavathi	Vijayawada	Deposit	11000.00	2024-01-16
108	Sush	Hyderabad	Deposit	9000.00	2024-01-14
NULL	NULL	NULL	NULL	NULL	NULL

Group By + Having + Order By

```
SELECT branch_name,
       AVG(amount) AS Avg_Amount
FROM bank_transactions
GROUP BY branch_name
HAVING AVG(amount) > 7000
ORDER BY Avg_Amount DESC;
```

Output:

branch_name	Avg_Amount
Vizag	9000.000000
Vijayawada	8000.000000

Where + Group By + Having

```
SELECT branch_name,
       SUM(amount) AS Total_Deposit
FROM bank_transactions
WHERE transaction_type='Deposit'
GROUP BY branch_name
HAVING SUM(amount) > 15000;
```

Output:

branch_name	Total_Deposit
Vijayawada	23000.00
Vizag	23000.00